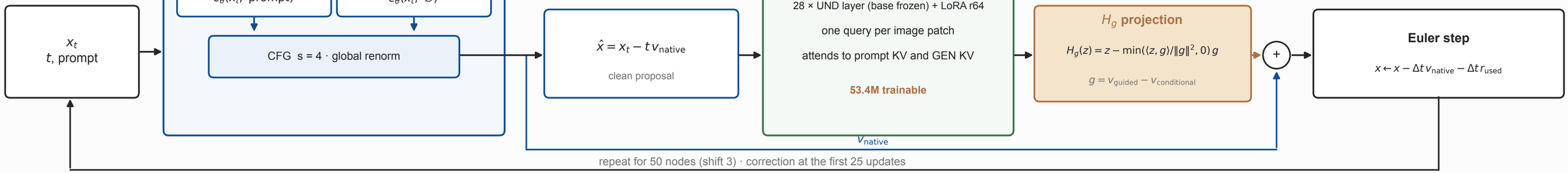
BAGEL · Frozen GEN + Spatial UND Velocity Correction

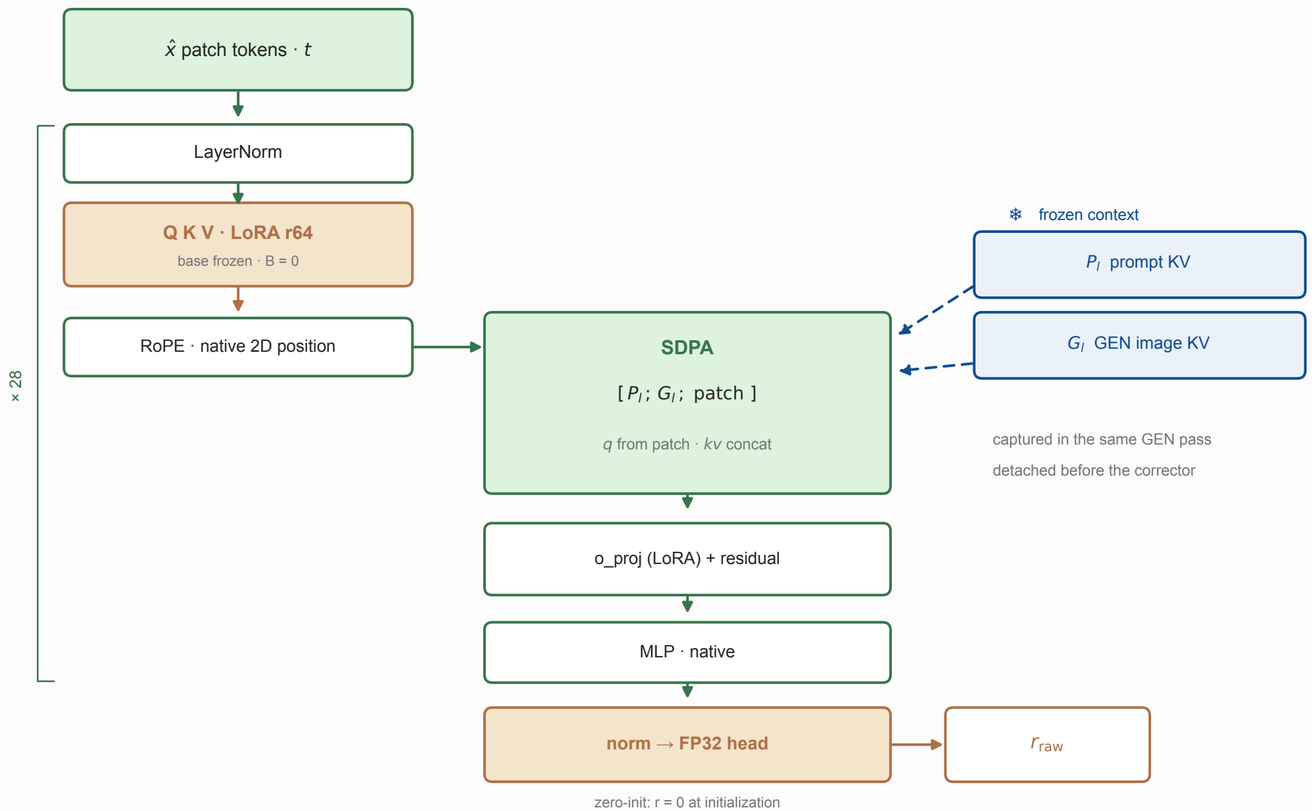
Method

❄ Frozen · Trainable · Correction path · Frozen KV context

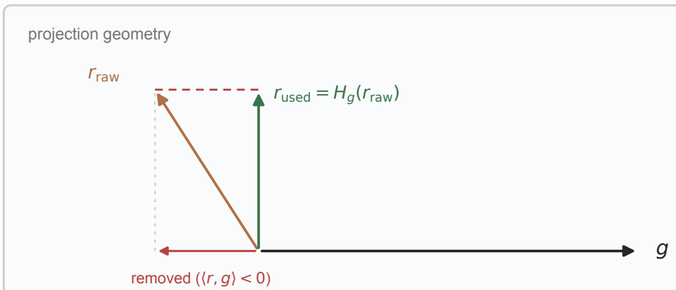
A Sampling step



B Corrector layer



C Guidance-safe objective



$g = v_{\text{guided}} - v_{\text{conditional}}$

$H_g(z) = z - \min(\langle z, g\rangle/\|g\|^2,\ 0)\,g$

$\text{target} = H_g(\varepsilon - x_0 - v_{\text{conditional}})$

$\text{loss} = \text{MSE}(r_{\text{raw}},\ \text{target})$

$r_{\text{used}} = H_g(r_{\text{raw}})$ at sampling

Trainable 53.4M · LoRA (28 layers) · input projection · FP32 head

❄ Frozen GEN (MMDiT) · UND base · prompt KV · GEN KV